Mr Michael Weiss

Network Service & Virtualization

Proof of Concept Linux virtual network Project

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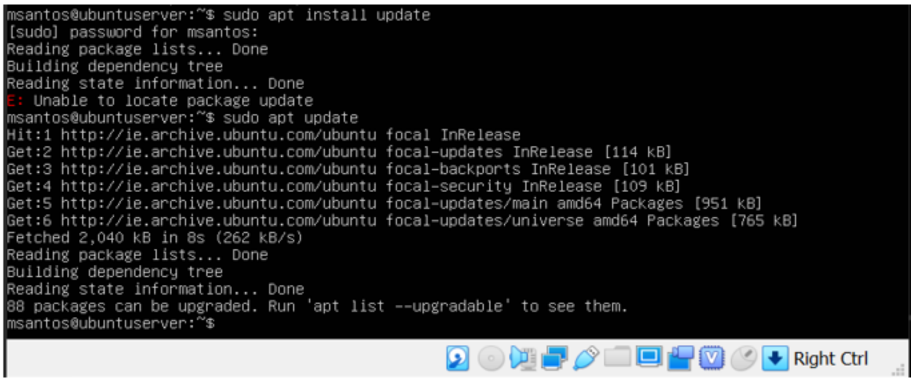
Summary

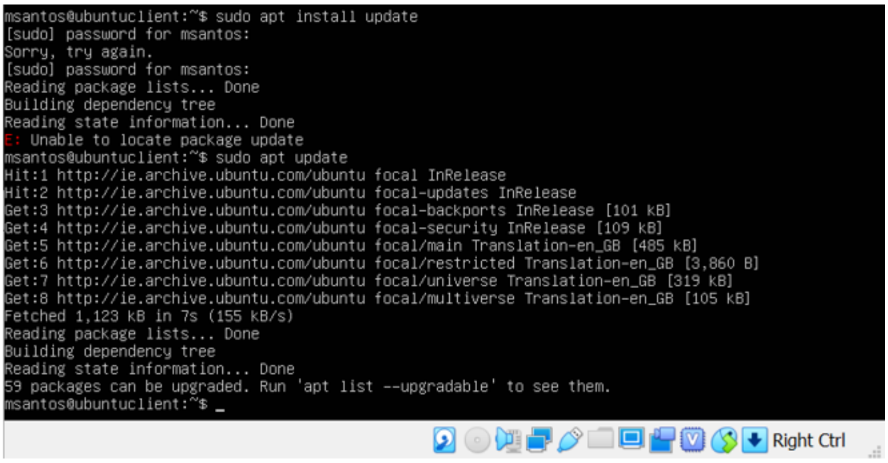
1- Obtain Linux updated and ping to the servers

Ubuntuserver update



```
msantos@ubuntuserver:~$ sudo apt install update
[sudo] password for msantos:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package update
msantos@ubuntuserver:~$ sudo apt update
Hit:1 http://ie.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://ie.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:3 http://ie.archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:4 http://ie.archive.ubuntu.com/ubuntu focal-security InRelease [109 kB]
Get:5 http://ie.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [951 kB]
Get:6 http://ie.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [765 kB]
Fetched 2,040 kB in 8s (262 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
88 packages can be upgraded. Run 'apt list --upgradable' to see them.
msantos@ubuntuserver:~$
```


Ubuntuclient update



```
msantos@ubuntuclient:~$ sudo apt install update
[sudo] password for msantos:
Sorry, try again.
[sudo] password for msantos:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package update
msantos@ubuntuclient:~$ sudo apt update
Hit:1 http://ie.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://ie.archive.ubuntu.com/ubuntu focal-updates InRelease
Get:3 http://ie.archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:4 http://ie.archive.ubuntu.com/ubuntu focal-security InRelease [109 kB]
Get:5 http://ie.archive.ubuntu.com/ubuntu focal/main Translation-en_GB [485 kB]
Get:6 http://ie.archive.ubuntu.com/ubuntu focal/restricted Translation-en_GB [3,860 B]
Get:7 http://ie.archive.ubuntu.com/ubuntu focal/universe Translation-en_GB [319 kB]
Get:8 http://ie.archive.ubuntu.com/ubuntu focal/multiverse Translation-en_GB [105 kB]
Fetched 1,123 kB in 7s (155 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
59 packages can be upgraded. Run 'apt list --upgradable' to see them.
msantos@ubuntuclient:~$ _
```

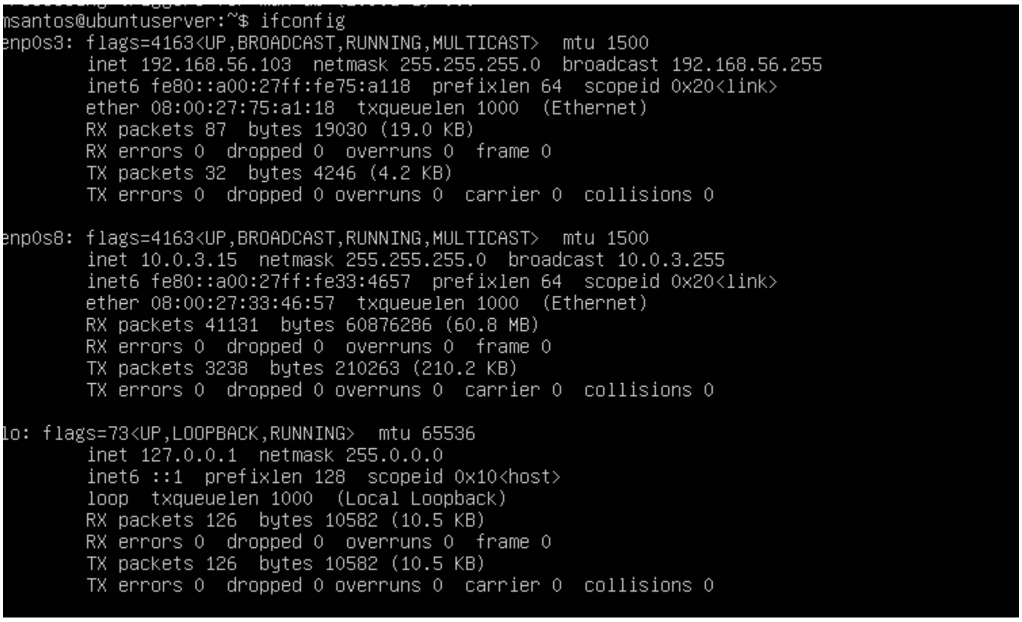
Ubuntuserver upgrade

```
E: Unable to locate package upgrade
msantos@ubuntuserver:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libevdev2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssh-server openssh-sftp-server openssl pciutils pollinate
  procps python3-apport python3-distupgrade python3-problem-report python3-software-properties
  python3-twisted python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen
  snapd software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
96 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 248 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Ubuntuclient upgrade

```
E: Unable to locate package upgrade
msantos@ubuntuclient:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libevdev2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssl pciutils pollinate procps python3-apport
  python3-distupgrade python3-problem-report python3-software-properties python3-twisted
  python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen snapd
  software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
94 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 247 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Ifconfig ubuntuserver

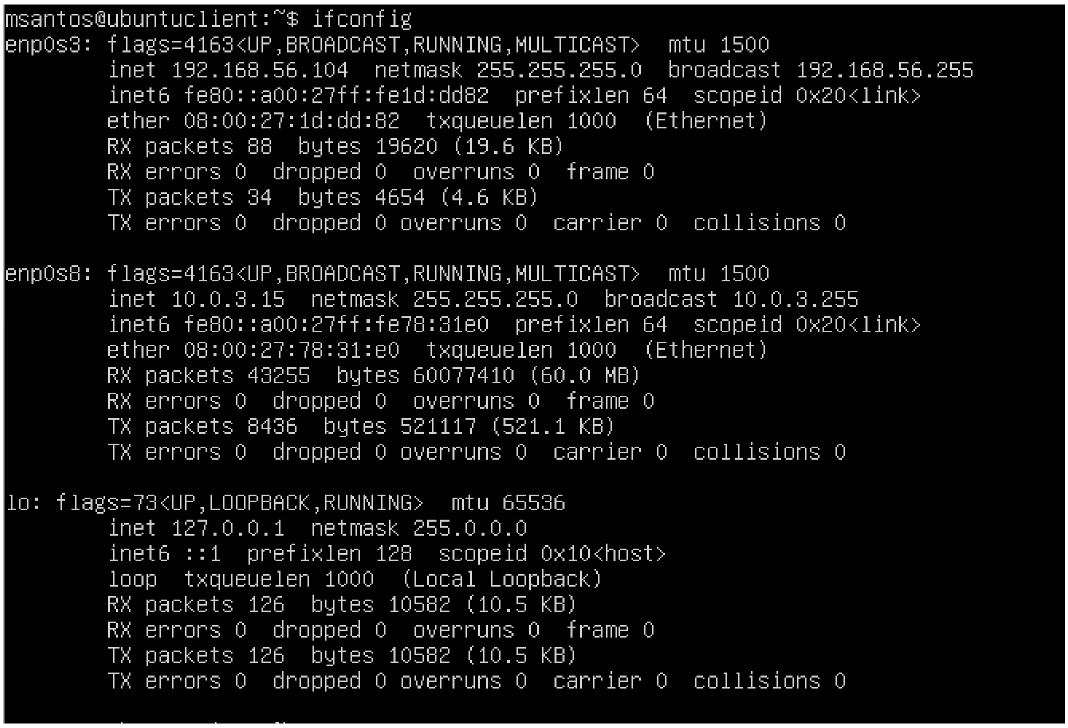


```
msantos@ubuntuserver:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.103  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::a00:27ff:fe75:a118  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:75:a1:18  txqueuelen 1000  (Ethernet)
        RX packets 87  bytes 19030 (19.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 32  bytes 4246 (4.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.3.15  netmask 255.255.255.0  broadcast 10.0.3.255
        inet6 fe80::a00:27ff:fe33:4657  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:33:46:57  txqueuelen 1000  (Ethernet)
        RX packets 41131  bytes 60876286 (60.8 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3238  bytes 210263 (210.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 126  bytes 10582 (10.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 126  bytes 10582 (10.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Ifconfig ubuntuclient



```
msantos@ubuntuclient:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.104  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::a00:27ff:fe1d:dd82  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:1d:dd:82  txqueuelen 1000  (Ethernet)
        RX packets 88  bytes 19620 (19.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 34  bytes 4654 (4.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.3.15  netmask 255.255.255.0  broadcast 10.0.3.255
        inet6 fe80::a00:27ff:fe78:31e0  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:78:31:e0  txqueuelen 1000  (Ethernet)
        RX packets 43255  bytes 60077410 (60.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8436  bytes 521117 (521.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 126  bytes 10582 (10.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 126  bytes 10582 (10.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Server pinging client

```
msantos@ubuntuserver:~$ ping 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=0.899 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=0.888 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=0.929 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=0.910 ms
64 bytes from 192.168.56.104: icmp_seq=5 ttl=64 time=0.904 ms
64 bytes from 192.168.56.104: icmp_seq=6 ttl=64 time=0.916 ms
64 bytes from 192.168.56.104: icmp_seq=7 ttl=64 time=0.610 ms
64 bytes from 192.168.56.104: icmp_seq=8 ttl=64 time=0.810 ms
^C
--- 192.168.56.104 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7011ms
rtt min/avg/max/mdev = 0.610/0.858/0.929/0.099 ms
msantos@ubuntuserver:~$ _
```

Install apache and display apache page.

Installing apache in the server

```
msantos@ubuntuserver:~$ sudo apt install apache2 -y_
```

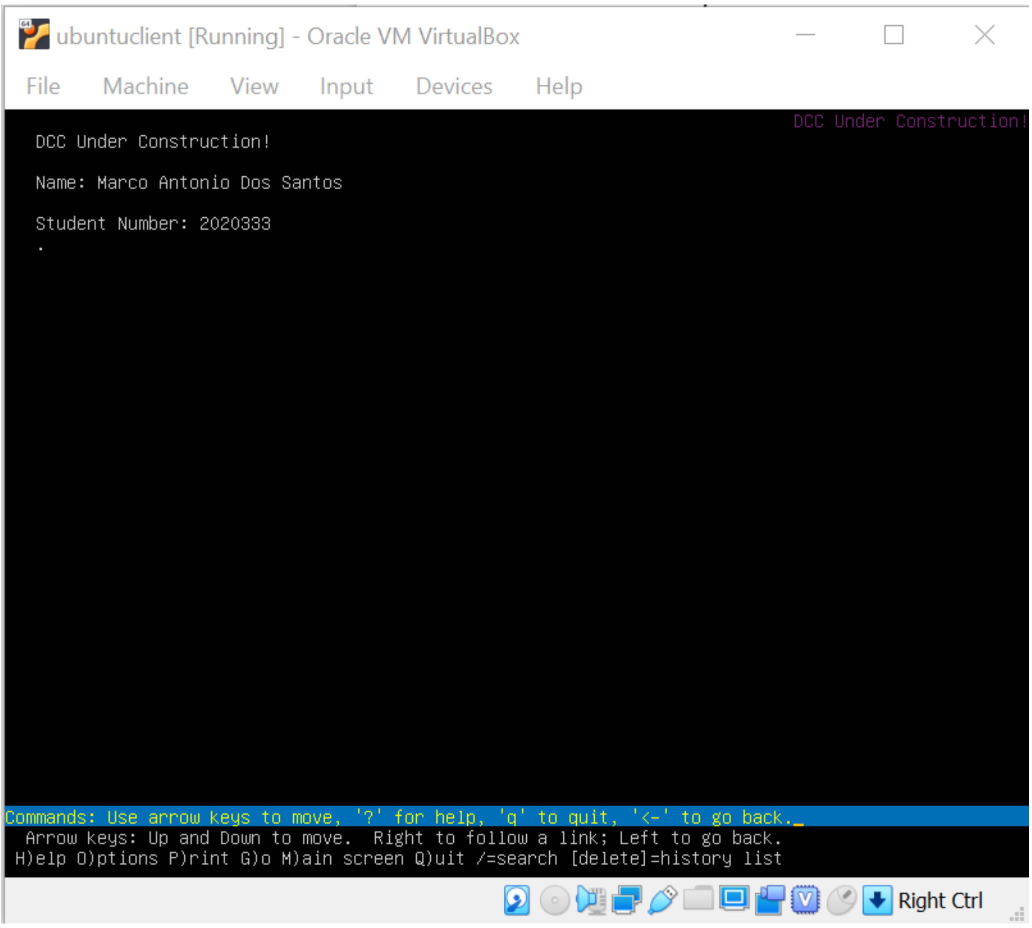
Installing lynx in the client

```
msantos@ubuntuclient:~$ sudo apt install lynx -y_
```

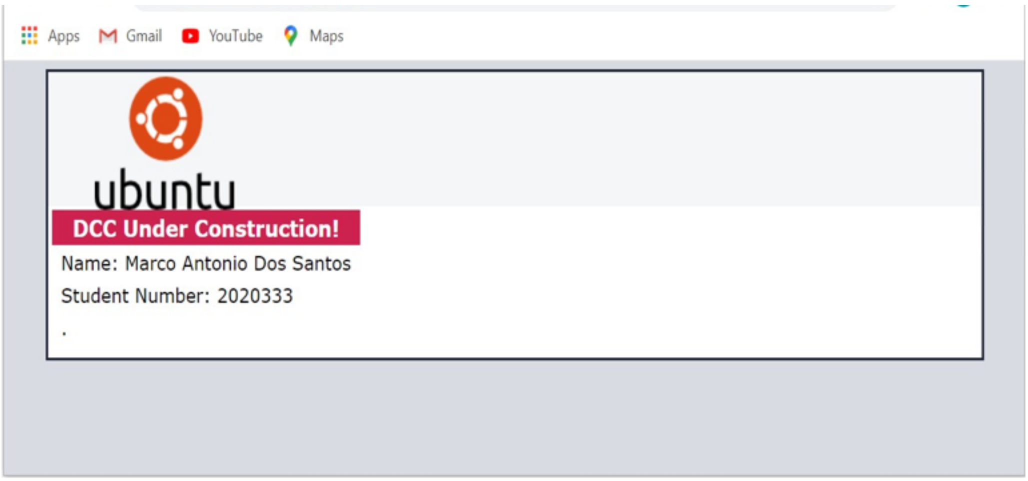
Client displaying apache page

```
msantos@ubuntuclient:~$ lynx www.apache.com
```

Displayed test to apache home page!



Host operating system to access to the Apache home page.



Host machine pinging server machine by command prompt.

```
C:\WINDOWS\system32>ping 192.168.56.103

Pinging 192.168.56.103 with 32 bytes of data:
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64
Reply from 192.168.56.103: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.56.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\WINDOWS\system32>
```

WIRESHARK

ICMP Traffic - host machine pinging server machine

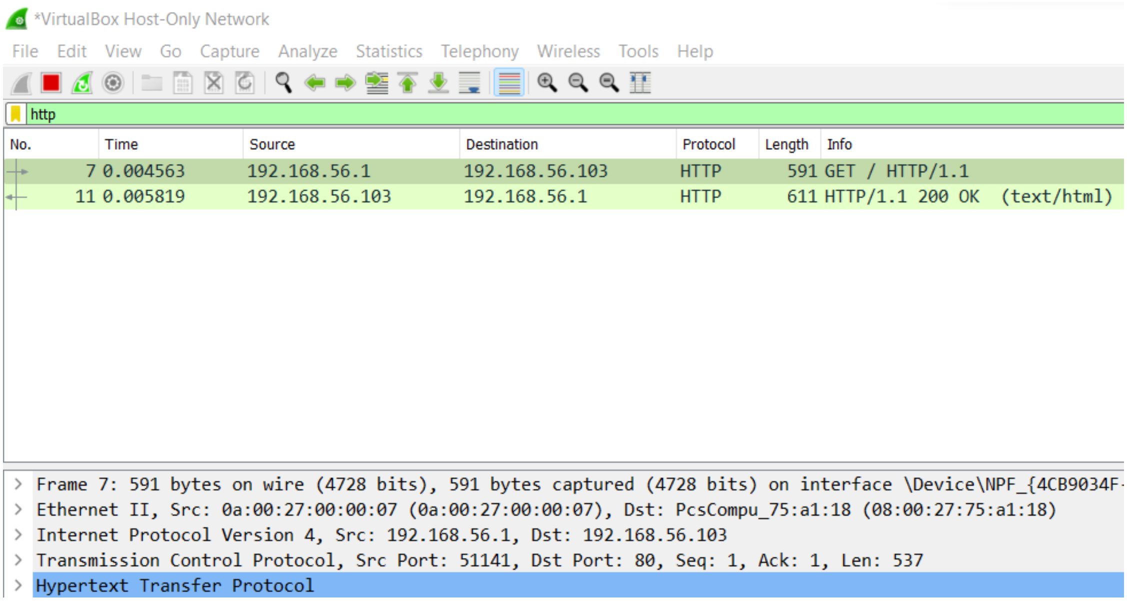
*VirtualBox Host-Only Network

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

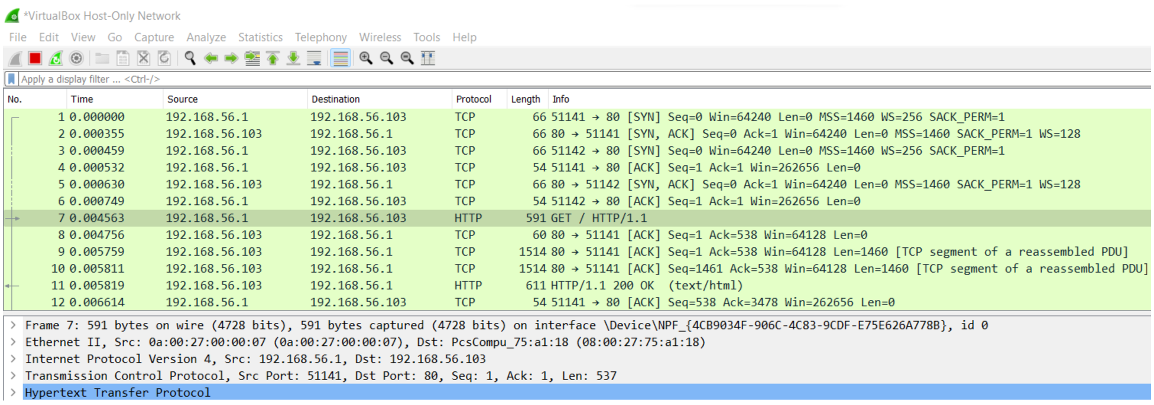
Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 2)
2	0.000276	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=17/4352, ttl=64 (request in 1)
3	1.002567	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=18/4608, ttl=128 (reply in 4)
4	1.003028	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=18/4608, ttl=64 (request in 3)
5	2.008815	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=19/4864, ttl=128 (reply in 6)
6	2.009178	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=19/4864, ttl=64 (request in 5)
7	3.013128	192.168.56.1	192.168.56.103	ICMP	74	Echo (ping) request id=0x0001, seq=20/5120, ttl=128 (reply in 8)
8	3.013389	192.168.56.103	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=20/5120, ttl=64 (request in 7)

http traffic after displaying apache webpage in google chrome browser

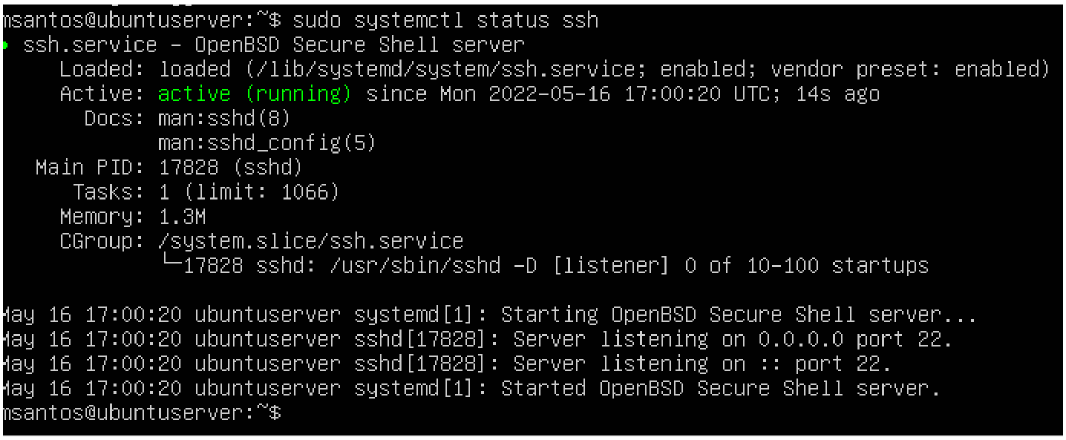


3 way handshake

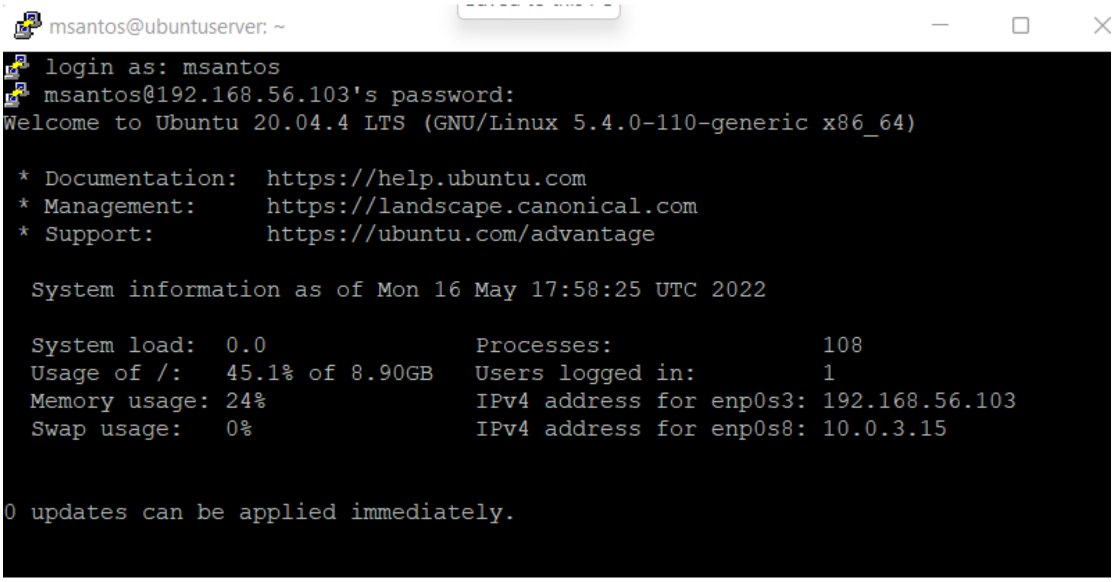


2-SSH logging and encrypted traffic

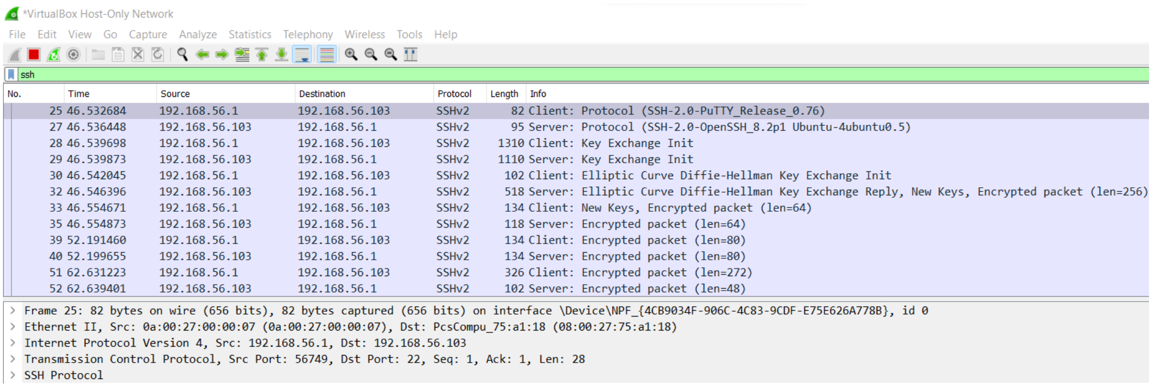
SSH installed



Accessing SSH



SSH traffic in Wireshark



No.	Time	Source	Destination	Protocol	Length	Info
25	46.532684	192.168.56.1	192.168.56.103	SSHv2	82	Client: Protocol (SSH-2.0-PuTTY_Release_0.76)
27	46.536448	192.168.56.103	192.168.56.1	SSHv2	95	Server: Protocol (SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5)
28	46.539698	192.168.56.1	192.168.56.103	SSHv2	1310	Client: Key Exchange Init
29	46.539873	192.168.56.103	192.168.56.1	SSHv2	1110	Server: Key Exchange Init
30	46.542045	192.168.56.1	192.168.56.103	SSHv2	102	Client: Elliptic Curve Diffie-Hellman Key Exchange Init
32	46.546396	192.168.56.103	192.168.56.1	SSHv2	518	Server: Elliptic Curve Diffie-Hellman Key Exchange Reply, New Keys, Encrypted packet (len=256)
33	46.554671	192.168.56.1	192.168.56.103	SSHv2	134	Client: New Keys, Encrypted packet (len=64)
35	46.554873	192.168.56.103	192.168.56.1	SSHv2	118	Server: Encrypted packet (len=64)
39	52.191460	192.168.56.1	192.168.56.103	SSHv2	134	Client: Encrypted packet (len=80)
40	52.199655	192.168.56.103	192.168.56.1	SSHv2	134	Server: Encrypted packet (len=80)
51	62.631223	192.168.56.1	192.168.56.103	SSHv2	326	Client: Encrypted packet (len=272)
52	62.639401	192.168.56.103	192.168.56.1	SSHv2	102	Server: Encrypted packet (len=48)

```
> Frame 25: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{4CB9034F-906C-4C83-9CDF-E75E626A778B}, id 0
> Ethernet II, Src: 0a:00:27:00:00:07 (0a:00:27:00:00:07), Dst: PcsCompu_75:a1:18 (08:00:27:75:a1:18)
> Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.103
> Transmission Control Protocol, Src Port: 56749, Dst Port: 22, Seq: 1, Ack: 1, Len: 28
> SSH Protocol
```

3-IP address and Hostname management

According to Daniel(2022):” Configuring a static IP can be difficult in Linux because it’s different based on the distro and version you’re using.”

Configuring server using Netplan.

ubuntuserver [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
  GNU nano 4.8                     /etc/netplan/00-installer-config.yaml
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [192.168.56.100/24, ]
      gateway4: 192.168.56.1
      nameservers:
          addresses: [192.168.56.1]_
    enp0s8:
      dhcp4: true
  version: 2
```

Sudo Netplan apply: new Ip address set

```
msantos@ubuntuserver:~$ sudo netplan apply
msantos@ubuntuserver:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.100  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::a00:27ff:fe22:e855  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:22:e8:55  txqueuelen 1000  (Ethernet)
        RX packets 44  bytes 7393 (7.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 36  bytes 4565 (4.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.3.15  netmask 255.255.255.0  broadcast 10.0.3.255
        inet6 fe80::a00:27ff:fe8a:cc93  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:8a:cc:93  txqueuelen 1000  (Ethernet)
        RX packets 232520  bytes 325854711 (325.8 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 42975  bytes 2625620 (2.6 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 140  bytes 11920 (11.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 140  bytes 11920 (11.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

msantos@ubuntuserver:~$ _
```

Changing ip address of Ubuntuclient using Netplan.

```
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [192.168.56.125/24, ]
      gateway4: 192.168.56.1
      nameservers:
          addresses: [192.168.56.1]
    enp0s8:
      dhcp4: true
  version: 2
```

Change was successful

```
msantos@ubuntuclient:~$ sudo netplan apply
msantos@ubuntuclient:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.125  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::a00:27ff:fe1d:dd82  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:1d:dd:82  txqueuelen 1000  (Ethernet)
        RX packets 890  bytes 136014 (136.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 201  bytes 32891 (32.8 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.3.15  netmask 255.255.255.0  broadcast 10.0.3.255
        inet6 fe80::a00:27ff:fe78:31e0  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:78:31:e0  txqueuelen 1000  (Ethernet)
        RX packets 50780  bytes 70580785 (70.5 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 10044  bytes 631708 (631.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 212  bytes 18388 (18.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 212  bytes 18388 (18.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

msantos@ubuntuclient:~$
```

Rename Hostname(s)

Ubuntuservers rename

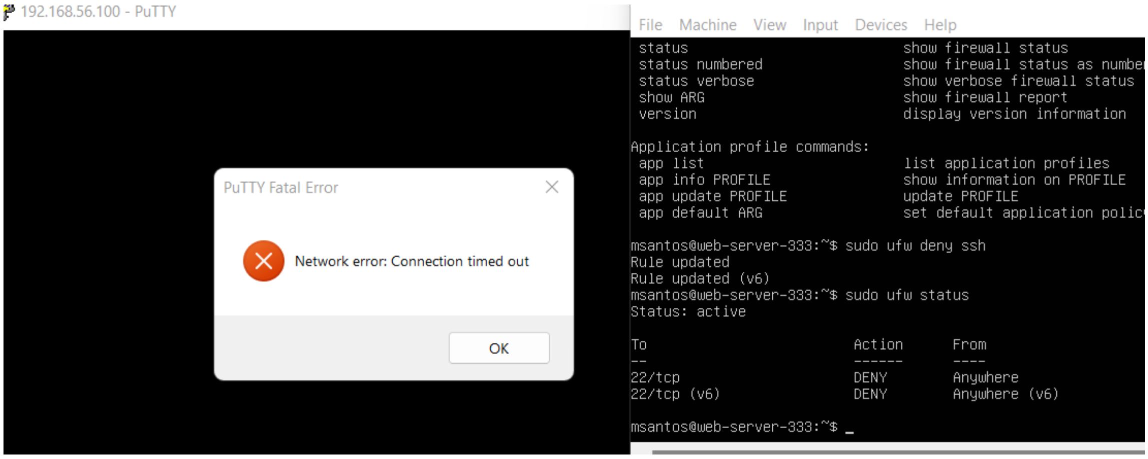
```
root@ubuntuserver:/home/msantos# cat /etc/hostname
ubuntuserver
root@ubuntuserver:/home/msantos# hostnamectl set-hostname web-server-333
root@ubuntuserver:/home/msantos# hostname
web-server-333
root@ubuntuserver:/home/msantos# hostnamectl
   Static hostname: web-server-333
         Icon name: computer-vm
           Chassis: vm
        Machine ID: 40621680576c401a8827ef4d7fbeab67
           Boot ID: f0ca8fbbb88f4b2193c1abc871dfe6c0
    Virtualization: oracle
  Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-65-generic
      Architecture: x86-64
root@ubuntuserver:/home/msantos# hostname
web-server-333
```

Ubuntuclient rename

```
root@ubuntuclient:/home/msantos# cat /etc/hostname
ubuntuclient
root@ubuntuclient:/home/msantos# hostnamectl set-hostname web-client-333
root@ubuntuclient:/home/msantos# hostname
web-client-333
root@ubuntuclient:/home/msantos# hostnamectl
   Static hostname: web-client-333
         Icon name: computer-vm
           Chassis: vm
        Machine ID: 9a0687241d3e4aecadb9047e9e74d4cc
           Boot ID: b87e49cf7ae84c60ab1eaa98ae091950
    Virtualization: oracle
  Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-72-generic
      Architecture: x86-64
root@ubuntuclient:/home/msantos# hostname
web-client-333
```

4-Firewall

Permission denied. Firewall is active but denying SSH.



Firewall is active. SSH access granted

The same process was repeated in UbuntuClient.

Reference:

Miessler,D.,2022.How to Set a Static IP Address in Linux .[Online] Available at: **https://danielmiessler.com/study/manually-set-ip-linux/** ,[Accessed 19 May 2022].